

SAHIL RAJESH WARKE

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PROFESSIONAL SUMMARY

Computer Engineering student and systems researcher specializing in **speculative parallel scheduling algorithms** and **memory-bounded probabilistic streaming systems**. Author of **HyperAgent** (patent-pending speculative P-DAG scheduling engine achieving 2.25x speedup with zero state contamination), **AegisStream** (streaming cyber-telemetry engine achieving 95.05% memory reduction via sub-linear sketches), and a published 276-page philosophical psychology work. Architect of **60 production web applications and simulation suites** deployed on GitHub Pages with an **8.24 CGPA** across 4 completed semesters, actively preparing for GATE CS/IT.

TECHNICAL SKILLS

Systems & Algorithms:	Speculative Parallel Scheduling, Probabilistic Streaming Sketches (CMS, HLL, Bloom Filter), Bounded Bipartite Trackers, Kahn's Waves, Copy-on-Write State Trees, LRU Caching
Languages & Core:	Python 3, JavaScript (ES6+), C/C++, HTML5, CSS3, SQL, Shell Scripting
Frontend & Graphics:	React.js, Vite, HTML5 Canvas API, SVG Vector Paths, Progressive Web Apps (PWA), Web Audio API
Tools & Infrastructure:	Git/GitHub, GitHub Pages, Google Cloud Platform (GCP), Linux/Windows, LocalStorage Engines

FLAGSHIP SYSTEMS RESEARCH & ENGINEERING SUITES (60 DEPLOYED APPS)

- 1. Distributed Systems & AI Scheduling: HyperAgent** ([Patent-Pending Architecture, IEEE Manuscript](#)) [Live Workbench](#)
 - Formulated agent workflows as Probabilistic DAGs (P-DAG) using Kahn's topological concurrency wave partitioning to bypass serial LLM token validation.
 - Implemented branch-isolated copy-on-write state trees with deterministic O(1) pointer rollback on mismatch, proving S_HyperAgent == S_Sequential across a formal 7/7 automated test suite.
 - Conducted cold-cache accuracy sweeps, establishing empirical break-even ($\alpha^* = 40\%$) and achieving 2.03x to 2.25x speedup over sequential execution with sub-millisecond LRU caching (p50 = 0.0018ms).
- 2. Streaming Algorithms & Cybersecurity: AegisStream** ([Sub-Linear Sketches, IEEE Manuscript](#)) [Live Telemetry Dashboard](#)
 - Engineered a memory-bounded streaming intrusion detector using Count-Min Sketch (conservative update), HyperLogLog arrays, and double-hashed Bloom filters, eliminating raw flow table bloat.
 - Invented a bounded bipartite lateral tracker replacing unbounded edge graphs with per-user HLL cardinality estimators, flatlining memory growth ($dM/dN \rightarrow 0$) with a 95.05% memory reduction at N = 500,000 events (3.60 MB vs 72.85 MB exact baseline).
 - Maintained steady throughput of 19,550 events/sec with 41.5 μ s median latency, achieving 100% precision across 12,000 benign flows and progressive multi-factor escalation across a 4-stage APT kill chain.
- 3. Symmetrical Game Systems & Forensic Simulations** ([Project L.I.Z.](#), [Fractured Self](#), [LinguaWar](#), [Pawnova](#)) [Live on GitHub Pages](#)
 - Architected symmetrical multi-perspective game loops driven by dual synchronized state machines (Detective vs. Killer) sharing a canonical mutable crime scene model with Bezier deduction corkboards.
 - Engineered an interrogation biometric engine with live ECG simulation, composure decay, and zero-asset procedural audio generation via the Web Audio API.
- 4. World Linguistics & Writing Systems Suite** ([18 Platforms: GreekFlow, HebrewFlow, KhmerFlow, ArmenianFlow, etc.](#)) [18 PWA Applications](#)
 - Developed 18 specialized linguistics platforms decoding ancient and modern scripts (abugidas, alphabets, logographs) across Greek, Hebrew, Khmer, Amharic, Cyrillic, Georgian, and Burmese.
 - Integrated dynamic SVG stroke order vector animations, IPA audio guides, Devanagari/Ge'ez ligatures, spaced-repetition drills, and offline PWA service workers.

PUBLISHED LITERARY WORK

"The Questions That Heal" ([Published Philosophical Psychology Work, 276 Pages](#)) [Internet Archive Public Access](#)
Authored a 276-page philosophical psychology work exploring memory, trauma recovery, emotional resilience, and self-inquiry through the journey of Professor Vinay across 10 structured therapeutic arcs. Preserved with permanent open-access indexing on Internet Archive.

EDUCATION & ACADEMIC PREPARATION

Bachelor of Engineering (B.E.) in Computer Engineering - MET's Institute of Engineering, SPPU, Pune **2024 - 2028 (Expected)**
Cumulative Grade Point Average (CGPA): **8.24 / 10.00** (Semesters I - IV completed) [Nashik, Maharashtra](#)
GATE CS/IT & Core Systems: Data Structures & Algorithms, Operating Systems, DBMS, Theory of Computation, Computer Networks, Discrete Mathematics.